

Byung-cheol Kang

Bacteriophage Research · Microbial Genomics · Workflow Engineering

I turn complex microbial and phage sequencing data into reproducible analysis systems, inspectable evidence, and research decisions.

CURRENT ROLE

Microiotix · Bioinformatician

2026.07 - PRESENT

Bacteriophage research, in silico design, and research database development and evaluation. Specific tools, database types, metrics, and outcomes are intentionally omitted until they are publicly verifiable.

CAREER SNAPSHOT

6+ years

Applied bioinformatics

5-person

Team led at Sanigen

4 profiles

16S · ITS · WGS · Hybrid

7 papers

SCI(E) publications

Past platform evidence. The integrated workflow and operational results in this portfolio describe work performed at Sanigen (2020.03 - 2025.10). They are not presented as Microiotix assets or current-company outcomes.

Integrated Microbial Genomics Platform

PROBLEM

Separate assay scripts, environments, quality checks, server paths, and reporting conventions made analysis difficult to reproduce and review consistently.

SYSTEM DESIGN

Entry point	One Snakemake dispatcher selects 16S, ITS, WGS, or Hybrid profiles.
Execution	Server-specific YAML configuration with Conda, Docker, or Apptainer and pinned BioContainers.
Quality	Input validation, raw-read QC, assembly/analysis checks, review gates, and retained logs.
Delivery	Custom Python/R processing, visualizations, HTML reports, and packaged outputs.

ACTIVE TOOL CHAINS

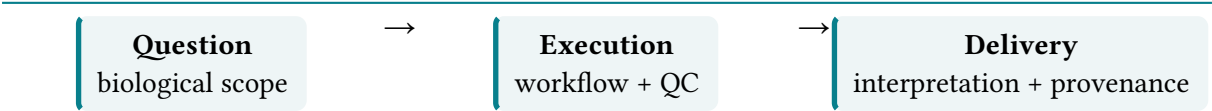
Profile	Core processing	QC / interpretation
16S / ITS	QIIME 2, DADA2, SILVA classifier, MAFFT, FastTree	FastQC, MultiQC, diversity metrics, custom R/Python reports
WGS	Trimmomatic, SPAdes, Bowtie2, samtools, Prokka	QUAST, BUSCO, Kraken2, Abricate/CARD/VFDB, eggNOG-mapper
Hybrid	NanoFilt, NanoPlot, Flye or Unicycler, BWA, Pilon, minimap2	QUAST, BUSCO, Prokka, CGView, BLASTn, Abricate, eggNOG-mapper

EVIDENCE BOUNDARY

Only tools connected to active Snakemake includes and rules are listed as active. Configured or documented but inactive components are not represented as production use.

Analysis Operations and Quality Assurance

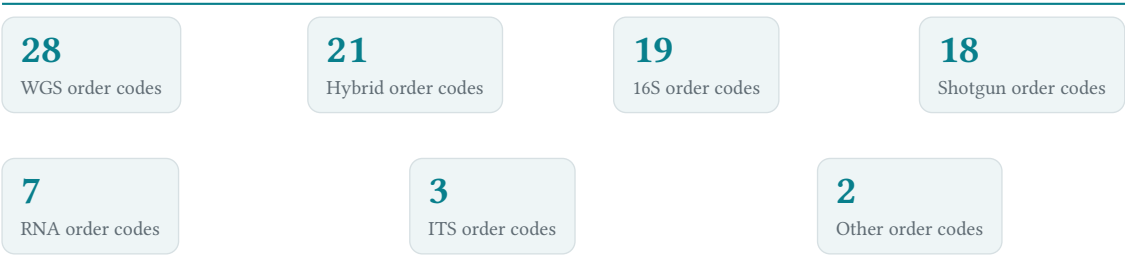
OPERATING MODEL



RESPONSIBILITIES AT SANIGEN

- Translated research and client questions into analysis scope, metadata requirements, and appropriate workflow profiles.
- Reviewed Q20/Q30, read depth, contiguity, assembly plausibility, taxonomy, annotation, and analysis-specific quality evidence before interpretation.
- Coordinated reruns, technical communication, report packaging, manuals, and training across a five-person team.
- Maintained reproducible environments and operational failure handling across Linux compute and storage infrastructure.

VERIFIED LOWER BOUND OF ARCHIVED WORK



These 98 distinct order codes are a conservative lower bound derived from archived communications. They are not sample counts, revenue, or an independently audited total workload.

From Genome Differences to Assay Candidates

COMPARATIVE-GENOMICS MARKER DISCOVERY

Objective. Identify a region that distinguishes a target group while remaining practical for downstream marker or assay development.

Approach. Curate complete genomes, compare whole-genome and pan-genome context, narrow candidate intervals, and review conservation, specificity, and downstream constraints.

Outcome. Prioritized an approximately **17 kb candidate region** for follow-up, with traceable reasoning from genome selection to practical assay constraints.

FOODBORNE-PATHOGEN NGS PANEL

- Contributed bioinformatics and comparative-genomics analysis to marker-oriented panel development and evaluation.
- Supported interpretation of NGS outputs across food and environmental sample contexts.
- Collaborative outcomes included two peer-reviewed papers in 2023 and one joint patent contribution in 2022.

BOUNDARIES

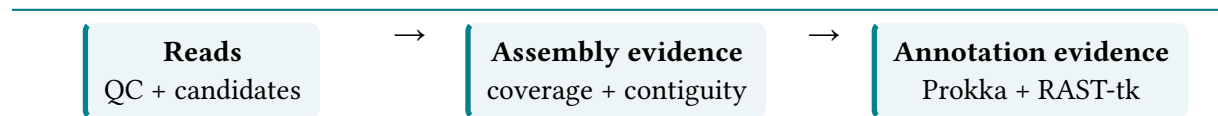
Computational prioritization does not establish diagnostic sensitivity or specificity. Publication and patent records document collaborative outcomes, not sole authorship or universal performance across unevaluated matrices.

SELECTED EVIDENCE

- **Frontiers in Microbiology** 14 (2023): agricultural wastewater pathogen panel.
- **Journal of Microbiology and Biotechnology** 33(1) (2023): fermented-food pathogen panel.
- Joint patent contribution (2022): NGS-based primer set and foodborne-bacteria detection method.

Bacteriophage Assembly and Annotation

WORKFLOW



CONTRIBUTION

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- Compared assembly candidates using contiguity, read support, and genome-scale plausibility.
 - Produced single-contig final assemblies across the assessed sample set.
 - Supplemented Prokka output with RAST-tk evidence rather than treating one annotation source as definitive.
 - Evaluated Oxford Nanopore and Nextflow-based processing as a separate portability and adoption track.

REVIEW PRINCIPLE

Assembly confidence and biological interpretation remain separate. A one-contig assembly is not automatically complete, circular, or correctly oriented; appropriate read-support and terminal-structure checks remain necessary.

TOOL AND METHOD EVIDENCE

Platform	Illumina; Oxford Nanopore evaluated in related work
Workflow evidence	Nextflow evaluated for portable ONT processing
Annotation	Prokka used; RAST-tk used as complementary evidence
Review	Coverage, contiguity, terminal structure, and biological plausibility

Phage Lifestyle Evidence Prototype

RESEARCH QUESTION

Can existing tools be combined into an end-to-end system that distinguishes lytic and lysogenic phages while preserving conflicting evidence and uncertainty?

EVIDENCE DESIGN

Quality context	CheckV completeness and contamination context
Annotation	Pharokka and Phold; integrase, repressor, excision, and lysis evidence
Classifiers	BACPHLIP and PhaTYP as independent prediction evidence
Decision layer	Agreement/confidence table with unresolved cases routed to manual review

HOST-RANGE EXTENSION

A related hypothesis-ranking layer considers LPS and outer-membrane gene variation together with phage receptor and annotation evidence. It prioritizes candidates for experiment; it does not prove adsorption, infection, or lytic activity.

PUBLICATION-GRADE NEXT STEPS

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- External, leakage-controlled benchmark with known lifestyle labels.
 - Comparison against individual tools and transparent simple baselines.
 - Sensitivity, specificity, MCC, calibration, and quality-stratified performance.
 - Ablation analysis for gene markers, classifiers, and quality context.
 - Experimental validation for host-range candidate claims.

CONTACT

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